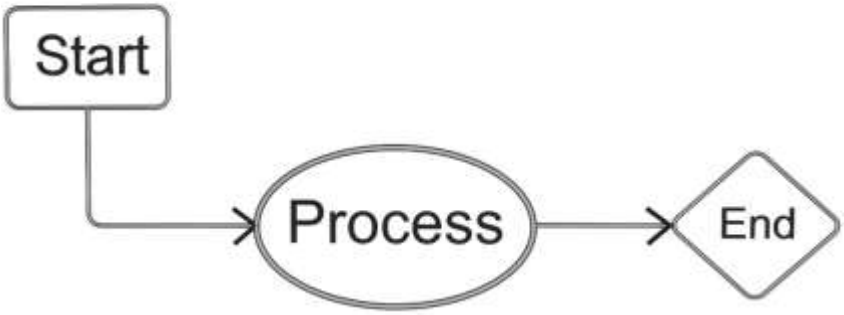
Process States ❁

In an operating system, a **process** is an instance of a program being executed. During its lifecycle, a process transitions between various states depending on its current activity. A process can only be in one state at any given time.

The most common process states are:

- **New**: This is the initial state of a process. A new process is one that has just been created by the operating system but hasn't yet been loaded into main memory.
- **Ready**: A process is in the ready state when it is waiting to be assigned to a CPU for execution. It's loaded in main memory and is ready to run, but the CPU is busy with another task.



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- **Running**: The process is in the running state when its instructions are actually being executed by the CPU. At any given moment, only one process can be in this state per CPU core.
- **Waiting (or Blocked)**: A process enters the waiting state when it is performing an action that requires it to wait for an event to occur. Common causes include waiting for user input, waiting for a file to be read, or waiting for an I/O operation to complete.
- **Terminated**: A process is in the terminated state after it has completed its execution. The operating system removes the process from the running queue, but its process control block (PCB) might be retained for a short time to provide status information.